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Assessment : 01

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1. What is Data Science? Explain its primary objective.

Data Science is an interdisciplinary field that uses statistics, mathematics, programming, and machine learning to extract useful insights from data.

Primary objective: To analyze data and convert it into meaningful information that supports better decision-making and problem-solving.

2. Why is Data Science important in today's digital world? Mention any four reasons.

1. Helps organizations make data-driven decisions.
2. Predicts future trends and customer behavior.
3. Improves business efficiency and productivity.
4. Supports innovation through AI and machine learning.

3. List any four benefits of Data Science in business and society.

1. Better decision-making.
2. Increased operational efficiency.
3. Improved customer satisfaction.
4. Enhanced healthcare and public services.

4. Mention four real-world applications of Data Science.

1. Healthcare (disease prediction)
2. Finance (fraud detection)
3. E-commerce (product recommendations)
4. Transportation (traffic and route optimization)

5. What are the different facets of data? Briefly explain each.

1. **Structured Data:** Organized in rows and columns (e.g., databases).
2. **Semi-structured Data:** Partially organized using tags (e.g., XML, JSON).
3. **Unstructured Data:** No fixed format (e.g., images, videos, emails).

4. **Metadata:** Data that describes other data (e.g., file size, creation date).

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Type	Description	Example
Structured	Stored in tables with fixed format	Student database
Semi-structured	Has tags but no fixed schema	XML, JSON
Unstructured	No predefined format	Images, videos, social media posts

7. What is Big Data? State its key characteristics (5 Vs).

Big Data refers to extremely large and complex datasets that cannot be processed using traditional tools.

5 Vs:

1. Volume
2. Velocity
3. Variety
4. Veracity
5. Value

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The **Big Data ecosystem** consists of tools and technologies used to collect, store, process, and analyze large datasets.

Technologies:

1. Hadoop
2. Spark
3. Hive
4. HBase

9. What are the major phases of the Data Science process?

1. Data Collection
2. Data Cleaning
3. Data Integration
4. Data Transformation
5. Data Analysis
6. Model Building
7. Model Evaluation
8. Deployment and Presentation

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of collecting data from different sources for analysis.

Sources:

1. Databases
2. Websites/APIs

11. Why is data collection considered an important step in the Data Science process?

Data collection is important because:

- It provides the foundation for analysis.
- Better-quality data leads to more accurate models and reliable results.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of detecting and correcting errors in data.

Techniques:

1. Removing duplicate records.
2. Handling missing values.
3. Correcting inconsistent data.
4. Removing outliers.

13. What is data integration? Why is it necessary in Data Science?

Data integration is the process of combining data from multiple sources into a single dataset.

Necessity:

- Improves data consistency.
- Provides a complete view for better analysis.
- Eliminates redundancy.
- Supports accurate decision-making.

14. Explain data transformation with suitable examples.

Data transformation is the process of converting data into a suitable format for analysis.

Examples:

- Converting text categories into numbers.
- Normalizing numerical values.
- Changing date formats (DD/MM/YYYY to YYYY-MM-DD).

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information and patterns.

Techniques:

1. Descriptive analysis
2. Diagnostic analysis
3. Predictive analysis
4. Prescriptive analysis

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Analytics	Purpose
Descriptive	Explains what happened.
Predictive	Predicts what may happen.
Prescriptive	Suggests what actions should be taken.

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Model building is the process of creating a machine learning or statistical model using data to make predictions or decisions.

Steps:

1. Select the algorithm.
2. Split the dataset into training and testing sets.
3. Train the model.
4. Test and validate the model.
5. Fine-tune if necessary.

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Model evaluation measures how well a model performs and ensures reliable predictions.

Metrics:

1. Accuracy
2. Precision
3. Recall
4. F1-Score

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings helps stakeholders understand insights and make informed decisions.

Tools/Techniques:

1. Bar Chart
2. Line Chart
3. Pie Chart
4. Tableau (or Power BI)

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

Data Science models are deployed by integrating them into software applications, websites, mobile apps, or cloud platforms so they can make real-time predictions.

Examples:

1. Recommendation systems (Netflix, Amazon)
2. Fraud detection in banking
3. Medical diagnosis systems
4. Spam email filtering